

IILM UNIVERSITY · GREATER NOIDA

INTERSHIP REPORT

Cybersecurity Analyst Intern

Identity and Access Management · Tata Consultancy Services

Raghav Bhujel

B.Tech. CSE (Cybersecurity) · Roll No. 25SCS1003000765

School of Computer Science and Engineering
Academic Year 2025–26

Candidate and internship details

Student Name

Raghav Bhujel

Programme

B.Tech. Computer Science and Engineering (Cybersecurity)

Campus

Knowledge Park II, Greater Noida

Internship Role

Cybersecurity Analyst Intern

Practice

Cybersecurity Consulting

Industry Case Study

ABC Healthcare — Healthcare Data Breach Prevention

Roll Number

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University

IILM University, Greater Noida

Host Organisation

Tata Consultancy Services

Domain

Identity and Access Management (IAM)

Client Engagement

TechCorp Enterprises — IAM Readiness Assessment

Academic Session

Academic Year 2025–29

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Overview of the assignment

Two complementary streams of work: an industry case on healthcare data protection, and a consulting assessment of TechCorp Enterprises’ readiness for IAM implementation.

ROLE Cybersecurity Analyst Intern	DOMAIN Identity and Access Management	PRACTICE Cybersecurity Consulting
PROJECT LEAD Priya, IAM Architect	BUSINESS ANALYSIS Ravi, IAM Business Analyst	CLIENT TechCorp Enterprises

Objectives of the internship

01

To develop a working command of Identity and Access Management as a core cybersecurity discipline, including RBAC, multi-factor authentication, and access governance.

02

To examine, through a healthcare case, how IAM prevents unauthorised access to sensitive records and supports regulatory compliance.

03

To assess the organisational factors — scale, user types, legacy systems, cloud posture, and user experience — that shape an enterprise IAM strategy.

04

To design IAM recommendations that align with business processes and objectives, rather than treating identity as a purely technical control.

05

To practise professional consulting methods: stakeholder briefing, structured assessment, and roadmap thinking within a cybersecurity engagement.

Identity and Access Management

IAM ensures that the right identities access the right resources, at the right time, for the right reasons — and that every access can be evidenced.

Identification

Unique digital identities for employees, contractors, partners, and customers, bound to authoritative HR and partner records.

Authentication

Proof of identity through multi-factor authentication and single sign-on, reducing password-only risk at the perimeter of every application.

Authorisation

Role-based access control and least privilege so that each identity receives only the permissions required to perform its duties.

Accountability

Complete audit trails, access reviews, and governance so that every attempt and action can be reconstructed, justified, and reported.

Why industry case studies matter

Case analysis is how IAM moves from component lists to professional judgement.

- They convert IAM theory — RBAC, MFA, and access governance — into operational judgement against real security failures.
- They show how the same components are adapted across industries with different regulatory burdens and risk appetites.
- They make outcomes visible: fewer unauthorised-access incidents, stronger compliance posture, and restored institutional trust.
- They supply transferable methods for investigating similar identity failures in future cybersecurity roles.
- They establish IAM as a foundational layer of modern cyber defence, not an optional IT add-on.



CASE STUDY I

ABC Healthcare

A national hospital chain confronting unauthorised access to patient records.

Recurring, unjustified access to clinical records threatened confidentiality, regulatory standing, and the sanctity of patients' personal information. Identity was not bound tightly enough to role and need-to-know; the organisation could neither justify access nor reconstruct it after the fact.

The chain initiated a comprehensive IAM programme. In healthcare, a data incident is not only a security event — it is a breach of clinical trust.

IAM implementation at ABC Healthcare

01

Access controls

IAM enforced rigid, role-based permissions. Only authorised healthcare professionals could open the patient records required for their clinical or administrative duties — need-to-know, not standing access.

02

Audit trails

Every access attempt and subsequent action was logged. The organisation could detect unusual patterns quickly and reconstruct events for incident response and regulatory evidence.

03

Governance

Access was tied to responsibilities and reviewed as roles changed. Standing privilege was reduced, closing the gap through which unauthorised viewing of records had previously occurred.

Outcomes of the IAM initiative

The programme demonstrated that proactive identity controls belong in the clinical risk register, not only in the IT catalogue.

- A significant reduction in unauthorised-access incidents across the hospital chain.
- Confidentiality of patient records restored and sustained as an operational property, not a policy statement.
- Continued alignment with healthcare data-protection and privacy regulations.
- A demonstrated case for proactive IAM investment in high-sensitivity industries.
- A professional lesson: in healthcare, identity controls are patient-safety and dignity controls, not merely IT hygiene.

TechCorp Enterprises

A digital-transformation programme in which the protection of digital assets became a central cybersecurity requirement.

MANDATE

Assess TechCorp's readiness for IAM implementation and recommend a strategy that scales with the organisation's evolving cybersecurity needs.

REQUIRED UNDERSTANDING

Organisational structure, existing systems, and business processes had to be read first. Only then could a tailored IAM solution address the client's specific challenges and aspirations — rather than a generic product stack.

Aligning IAM with business processes

Identity controls that ignore how the organisation works become shelfware.

01 Stakeholder collaboration

Business owners, application teams, HR, and security must describe real workflows before entitlements are modelled. IAM that is designed in isolation fails at go-live.

02 Customisation

A one-size-fits-all suite rarely fits. Controls, approval chains, and role catalogues should be tailored to TechCorp's processes so that security increases efficiency rather than adding delay.

03 Scalability

The design must absorb growth in users, applications, and partners without a redesign. Joiner–mover–leaver processes are the true test of whether identity operations will scale.

04 Integration

IAM has to connect cleanly with directories, HR systems, cloud platforms, and line-of-business applications, or it becomes another silo that the business learns to work around.

05 User-centric design

Access request, authentication, and recertification should be intelligible to non-specialists. Friction that is unexplained is treated as a defect and is eventually ignored.

Frameworks, controls and standards

RBAC

Role-based access control as the primary authorisation model.

MFA

Multi-factor authentication to raise assurance beyond passwords.

Least privilege

Need-to-know entitlements; standing access kept to the minimum.

Access governance

Periodic reviews, recertification, and joiner–mover–leaver control.

Audit logging

Forensic-grade trails of attempts, grants, and administrative actions.

HIPAA

Healthcare privacy and security obligations in the ABC case.

PCI DSS

Comparative payments-security baseline for regulated IAM.

Lifecycle

Identity created, changed, and retired in step with employment and contracts.

Key learnings

- 01 IAM is simultaneously a technical control plane and a business-design problem; tools without process do not produce security.
- 02 Industry cases are how RBAC, MFA, and governance become judgement, not definitions committed to memory.
- 03 In healthcare, a failed identity control is a privacy incident and a breach of patient dignity — the ABC case made that concrete.
- 04 Consulting begins with structure, systems, and process. Architecture follows the organisation; it does not precede it.
- 05 User experience is a security property. Controls that are painful are circumvented; controls that are fluent are used.
- 06 Auditability is a first-class outcome. If access cannot be evidenced, it cannot be defended to a regulator or a court.

Conclusion

The internship established a working command of Identity and Access Management as practised in cybersecurity consulting. Through the ABC Healthcare case and the TechCorp readiness assessment, I learned to evaluate IAM not as a product purchase but as a tailored control system — sized to the organisation, bound to compliance, integrated with legacy and cloud estates, and designed around the people who must use it.

The experience confirms IAM as a foundational discipline for a career in cybersecurity, particularly in regulated and high-trust environments where identity failure is a business, privacy, and safety failure at once.

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END OF PRESENTATION

Thank you

Questions are invited

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